

Volodymyr Hryshko

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Łódź, Poland

Incoming **third year** Computer Science student with hands-on backend development experience in raw **Java**, **Spring Boot**, Python building secure, layered, and scalable systems. Fluent **English communicator (C1)** with a structured, detail-oriented approach to software engineering — adaptable and quick to learn new technologies, frameworks, and concepts.

EDUCATION

Łódź University of Technology

Łódź, Poland

International Faculty of Engineering, Computer Science (Incoming 5th Semester), Full Time · 2024 – present

Medium of Instruction – **English**, GPA: **4.01**

Expected Graduation: **May 2028**

Relevant courses:

Algorithms & Data Structures, Java Fundamentals, Databases, Software Engineering, Computer Architecture, Interactive Web Applications (Spring Boot)

TECHNICAL SKILLS

Languages: Java, Python, C

Frameworks & Libraries: Spring Boot, FastAPI, LangChain

Testing Frameworks: JUnit 5, Pytest

Concepts & Methodologies: OOP, REST, ORM, Unit Testing, Test-Driven Development, Agile Framework, Scrum, Embeddings, APIs

Developer Tools: Git, Linux (Ubuntu), CLI, Docker

Data & Infrastructure: PostgreSQL, ChromaDB, Vector DBs

AI: AI CLI tools, Claude Code CLI

PROJECTS

Universal University App | *Demo Video* | [Source code](#)

Designing and building a secure three-tier university management system meant to be universally applicable by any educational institutions, enabling students to view enrolled subjects and grades, and teachers to seamlessly create subjects and assign grades.

- Implemented a stateless JWT-based authentication and role-based authorization layer with Spring Security, including custom authentication filters, token provider, and user-principal modeling. Developed persistence layer using JPA/Hibernate with relational mappings across users, roles, subjects, enrollments, and grades, backed by PostgreSQL running in Docker.
- Built RESTful API endpoints following REST conventions, with BCrypt password hashing and CORS configuration for a decoupled frontend.

Tech Stack: Java 17, Spring Boot 4, Spring Security, Hibernate, PostgreSQL, Docker, Angular, TypeScript, Maven, Git

Fraud Detection RAG Pipeline (F-RAG) | *Demo Video* | [Source Code](#)

An application based on Retrieval Augmented Generation (RAG) technique that ingests financial documents and answers fraud-related queries using semantic search and a locally-run LLM, with a real-time citation interface.

- Built a full-stack Retrieval-Augmented Generation (RAG) pipeline that ingests PDF/DOCX documents, chunks and embeds them into a vector database, and retrieves semantically relevant passages in response to natural language queries.
- Integrated a locally-hosted Llama 3.2 LLM via Ollama to generate grounded, citation-backed answers from retrieved context — with a dual-mode toggle between raw retrieval and LLM generation.
- Developed a Tauri desktop application with a React/TypeScript frontend displaying real-time findings, source citations, and document management, backed by a FastAPI REST API.

Tech Stack: Python, FastAPI, uvicorn, ChromaDB, sentence-transformers (all-MiniLM-L6-v2), Llama 3.2 via Ollama (local, no API cost), LangChain text splitters, pytest

SQL Database Engine in Java | *Demo Video* | [Source Code](#)

SQL Database Engine – Java, Gradle, Jackson

- Designed and built a relational database engine from scratch in Java, implementing a hand-written SQL parser/tokenizer and an interactive CLI supporting CREATE, INSERT, SELECT, UPDATE, and DELETE with WHERE-clause filtering
- Applied the Strategy pattern to model comparison operators (=, !=, <, >, IN, BETWEEN) as pluggable classes, and a double-dispatch execution model to route parsed queries to their handlers
- Implemented a JSON-based persistence layer using Jackson to serialize/deserialize tables to disk, decoupling storage from query execution
- Built custom exception types for structured SQL syntax and database error handling

LANGUAGES

Polish — A2, spoken · English — C1, written and spoken · Ukrainian — native

CERTIFICATES

2024 Duolingo English Test (DET): **135 — C1 (CEFR)**

2024 National Multi-Subject Test (English): **200/200**

2023 IELTS: Band 6.5 — B2 (CEFR)